Current Status and Optimization of Long-Distance Coach Services under the Impact of High-Speed Rail

— A Case Study of the Yangtze River Delta

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Abstract: Using network indicators such as centrality and connectivity, together with disaggregate models, structural equation modeling (SEM) and binary logit regression, this paper examines the operating characteristics, strengths and weaknesses of long-distance coach services in the Yangtze River Delta (YRD) under the impact of high-speed rail (HSR), and analyzes passengers' preferences for travel modes. The results show that the YRD coach and HSR passenger networks share two common features, a distance-decay effect and a spatial agglomeration effect, and differ in two respects: the road passenger network has tighter internal connections, while the rail passenger network has shorter average travel times. The flexible and convenient location of coach stations and the frequent, varied schedules allow coaches to retain a certain market share among older passengers, but comfort and efficiency lead more than half of respondents to prefer HSR. This study aims to comprehensively assess the current situation of long-distance coaches in the new era, explore pathways for the transformation of scheduled road passenger services, and optimize the operating ecosystem of road passenger transport.

Keywords: Yangtze River Delta; long-distance coach; high-speed rail; transport network characteristics; travel intention

1. Introduction

In recent years, the improvement and spread of rail transit networks, which combine comfort, safety and high speed, have dealt a major blow to traditional long-distance road passenger transport. In 2018, national railways carried 3.375 billion passengers, up 9.4% year on year, and passenger-kilometers grew by 5.1%. Meanwhile, road passenger transport in China carried 13.672 billion commercial passengers, down 6.2%, with passenger-kilometers falling by 5.0%[1]. As long-distance coach services face a sharp loss of market share, how to re-assess their current situation comprehensively and explore pathways for the transformation of scheduled road passenger services has become a pressing problem.

Regarding the organizational characteristics and spatial structure of HSR and long-distance coaches, Yu Wenjia et al. studied the accessibility of the road passenger network in Jiangxi Province and found that its connection centers consist of stations with large passenger flows in each prefecture-level city[2]; Wang et al.[3] used departure frequencies between coach stations and railway stations in 361 Chinese cities to measure urban centrality and inter-city connectivity, and applied the modularity method[4] combined with PageRank to calculate the importance of each community within the overall transport network.

Regarding the competition between long-distance coaches and HSR, Zhou Li[5], from a multi-dimensional perspective and taking the Chongqing–Wanzhou corridor as an example, quantitatively analyzed the impact of HSR opening on road passenger transport; Niu Qiang[6] analyzed differences between HSR and coach services in operating environment, passenger-flow characteristics and changes in demand, and conducted an empirical analysis of the impact of HSR on road passenger

transport.

Overall, existing studies use a variety of methods to analyze the impact of HSR on long-distance coaches, but analyses of coach operations at the regional meso-scale are lacking, and few studies compare the characteristics of the supply side and the demand side together. This study covers 27 cities in the YRD metropolitan area and analyzes both current operations and passengers' travel preferences, offering a relatively comprehensive perspective.

2. Data Sources and Methods

2.1 Data Sources

This study uses route and schedule data for coaches and HSR between 27 YRD cities, as well as data from a YRD passenger travel questionnaire. The HSR and coach route and schedule data come from chepiao100.com, a website with relatively complete information on domestic coach routes, and were scraped with Python, totaling 50,848 records; the questionnaire data come from an online survey designed and distributed by our group, totaling 405 responses. Samples of the scraped data and the questionnaire are provided in Appendices 1 and 2.

2.2 Research Methods

The technical roadmap of this study is shown in Figure 1: first, OD information for HSR and coach services was scraped, and urban centrality and connectivity were used to analyze the characteristics of the road and HSR passenger networks; operations of coaches and HSR were then analyzed using schedule counts and related information. The questionnaire data were mainly analyzed with structural equation modeling and binary logistic regression to obtain evaluations of and suggestions for coaches and HSR. Finally, by comparing the results of the schedule data analysis with public perceptions, we propose recommendations for optimizing long-distance coach operations in the HSR era.

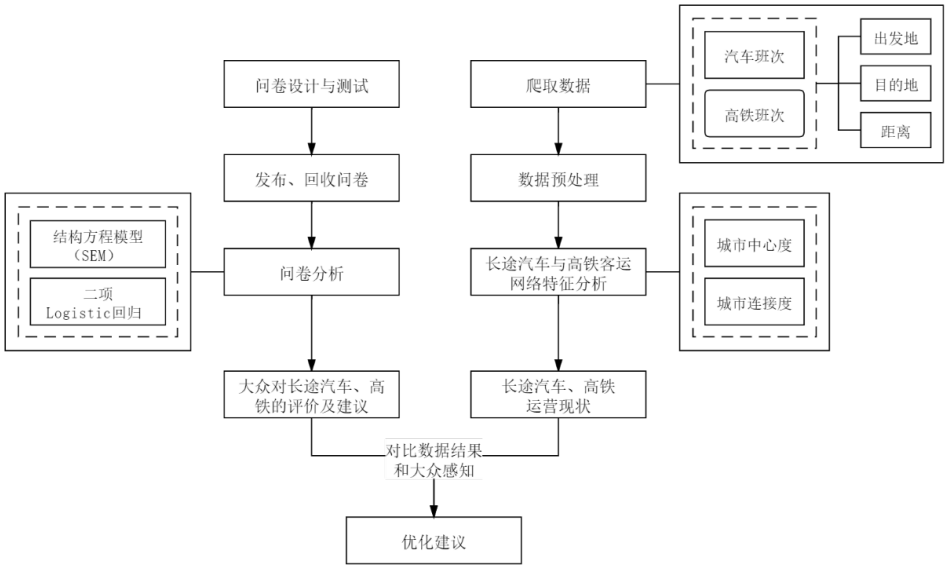


Figure 1 Technical roadmap

2.2.1 Network Structure Analysis of Road Passenger Transport and HSR

(1) Centrality

The centrality index proposed by Limtanakool et al. is used in this study to measure the hierarchical position of each YRD city in the HSR or coach network; its formula is as follows:

$$PI_i = \frac{T_i}{\left(\sum_{j=1}^{n} T_j / n\right)}$$

where T_I and Tj denote the total number of coach (or HSR) services connecting city i and city j; n is the number of YRD cities covered by coach (or HSR) services. A PIi greater than 1 indicates that the city holds a dominant position in the coach (or HSR) network.

(2) Connectivity

The connectivity between a pair of cities can be expressed by the Relative Strength of Links (RSL); its formula is as follows:

$$RSL_{ij} = \frac{t_{ij}}{\sum_{i=1}^{n} \sum_{j=1, j \neq i}^{n} t_{ij}}$$

where t_{ij} is the total number of coach (or HSR) services operating between the city pair, with $i \neq j$.

2.2.2 Disaggregate Model

A disaggregate model is built on individual survey data: the probability that an individual chooses a given travel mode is calculated from individual travel data, and the choices of many sampled individuals are then extended to obtain the travel-mode preferences of an urban area or region.[7] The theoretical basis of the disaggregate model is utility maximization theory, and it assumes no internal competition within each travel mode[8]. This paper uses a disaggregate model to explore passengers' preferences between HSR and coach travel.

2.2.3 Structural Equation Modeling (SEM)

Structural equation modeling (SEM) is a multivariate statistical technique that combines factor analysis and path analysis; it estimates the parameters of a structural model by constructing a system of latent variables and calculating their parameters. Our group used SmartPLS to solve for the latent variables of the SEM. The latent variable system and its indicators are shown in Table 1 below:

Table 1 Latent variable system

Latent variable	Meaning
Tangibles	The perceivable part of the service, such as the facilities used to provide transport services
Reliability	The service provider delivers the promised service accurately and without error
Responsiveness	The ability to respond, i.e. being ready at all times to provide passengers with quick and effective service
Assurance	Staff's good service attitude and competence, which strengthen passengers' confidence in and sense of security about the company's service quality
Empathy	The company and its staff put themselves in passengers' shoes and strive to meet passengers' needs

2.2.4 Binary Logit Regression Model

The binary logit regression model, also called the BL (Binary Logit) model, can be used when there are exactly two alternatives, A_1 and A_2, and the decision-maker chooses one of A_1 and A_2. Its expression is:

$$P_{1i}=\frac{exp^{(\lambda V_{1i})}}{exp^{(\lambda V_{1i})}+exp^{(\lambda V_{2i})}}$$

$$P_{2i}=1-P_{1i}=\frac{exp^{(\lambda V_{2i})}}{exp^{(\lambda V_{1i})}+exp^{(\lambda V_{2i})}}$$

where P_{in} is the probability that traveler n chooses alternative i, i = 1, 2; V_{in} is the deterministic utility of alternative i for traveler n.

3. Results

3.1 Network Characteristics of Long-Distance Coach and HSR Passenger Transport

Based on the number of HSR and coach services between the 729 city pairs of the 27 YRD cities, we calculated the centrality of each city and the connectivity of each city pair, with the results shown in Figures 2-1 and 2-2:

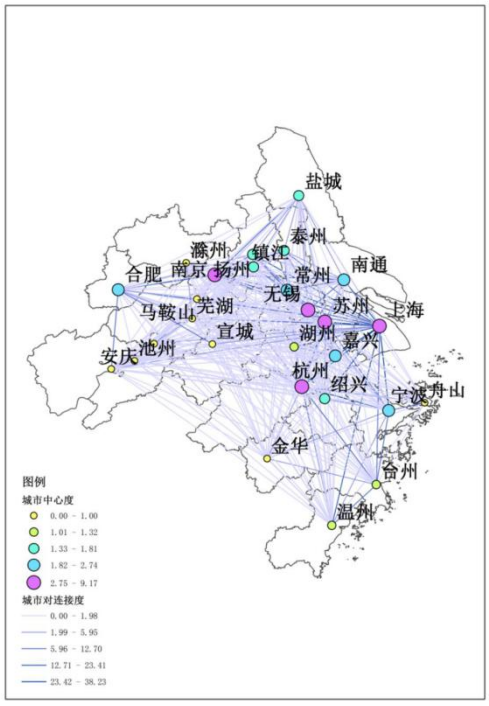


Figure 2-1 Centrality and connectivity of the coach network

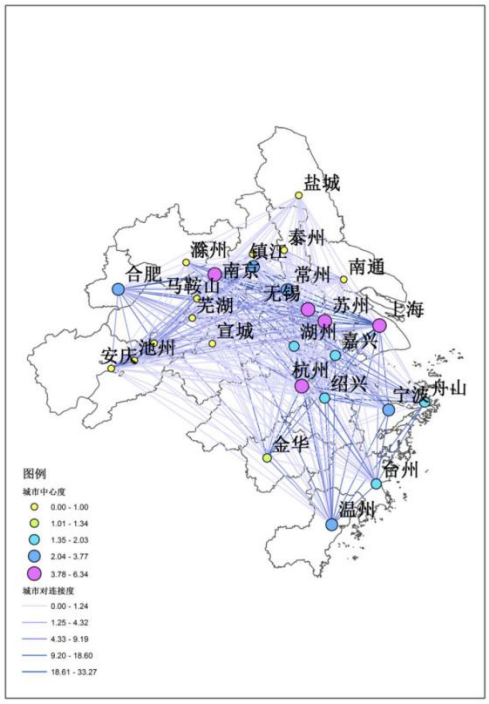


Figure 2-2 Centrality and connectivity of the HSR network

Based on the network structure analysis in Figures 2-1 and 2-2, the similarities and differences between the HSR and coach passenger networks are as follows:

(1) Similarity 1: A distance-decay effect

As two typical forms of ground transport, inter-city coaches and HSR are both constrained by geographic proximity, showing a spatial distance-decay effect. Figure 3 shows that within 600 km, the number of coach and HSR services first rises and then falls as distance increases. Beyond 500 km, coach and rail services each account for less than 1.0% of the total, indicating that both modes are concentrated on short- and medium-distance passenger transport. Meanwhile, Figure 4 shows that the distributions of coach and HSR services over the 200–300 km range are very similar, indicating very intense competition between the two modes.

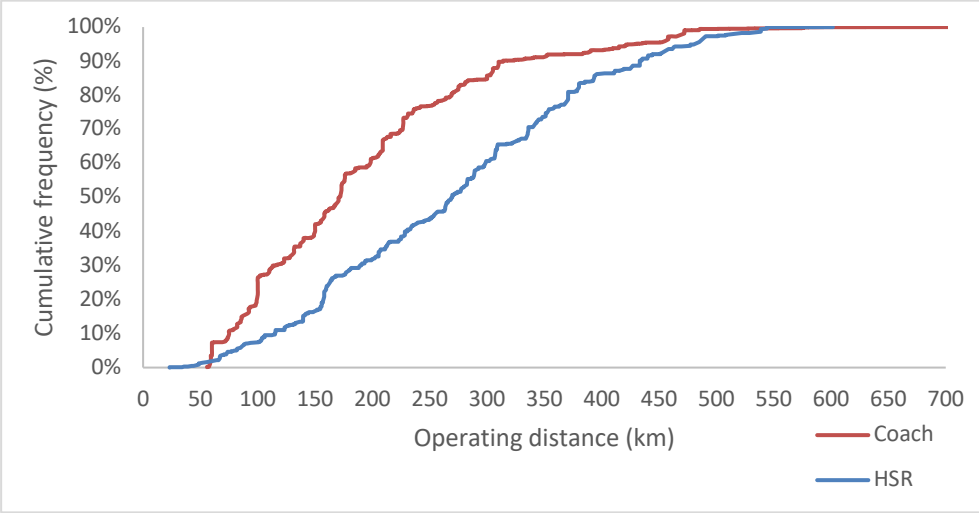


Figure 3 Cumulative distribution of operating distance for coaches and HSR

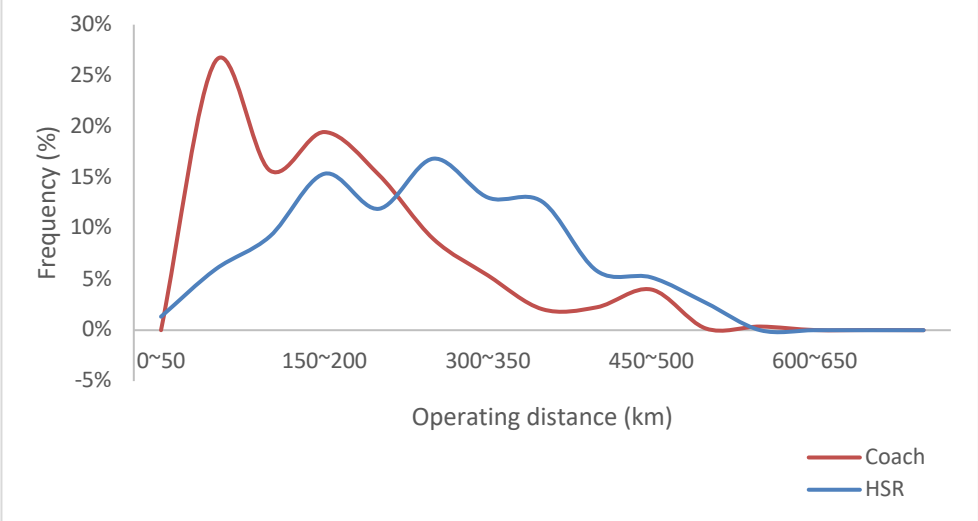


Figure 4 Distribution of operating distance for coaches and HSR

(2) Similarity 2: Spatial agglomeration and route preference

All cities and city pairs in the road and HSR networks were divided into five tiers according to urban centrality and connectivity, respectively. The top two tiers include fewer than 10% of city pairs, yet the coach and train services operating between them account for more than half of all services. This shows that both road and HSR passenger transport are concentrated in a few major

node cities and distributed along the main trunk cities.

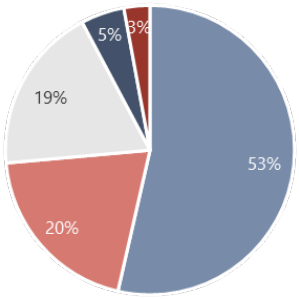


Figure 5-1 Share of coach departures

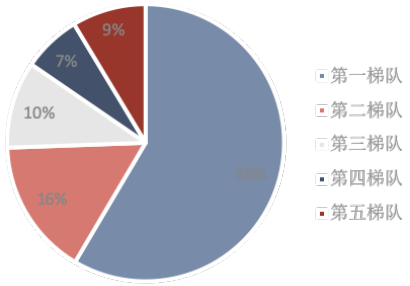


Figure 5-2 Share of HSR departures

(3) Difference 1: The long-distance coach network has tighter internal connections

The internal connections of the road passenger network are tighter than those of the HSR network, mainly in terms of the degree of distance decay and the distance between strongly connected city pairs. First, distance decay is more pronounced for road passenger transport. Figure 6-1 shows that in the 100–200 km range, the distribution frequency of coach services is higher than that of HSR; the gap in cumulative distribution frequency between the two modes is largest at around 200 km; beyond 200 km, the cumulative frequency of HSR begins to exceed that of coaches, and the gap gradually narrows as distance increases. This is consistent with previous findings that road passenger transport is more competitive over short distances while HSR is more competitive over medium distances. As for strongly connected cities, the most strongly connected city pairs in the road network are mostly within 200 km of each other, especially provincial capitals or large cities paired with surrounding satellite cities or nearby important cities, such as Wuxi–Suzhou, Hangzhou–Shaoxing and Hangzhou–Jiaxing, whereas the most strongly connected city pairs in the HSR network span greater distances, such as Shanghai–Nanjing and Nanjing–Suzhou.Figure

Table 2 Centrality of the coach and HSR networks: Top 10

Coach TOP 10	Centrality value	HSR TOP 10	Centrality value
Wuxi–Suzhou	38.22751323	Shanghai–Nanjing	33.27157635
Shanghai–Suzhou	35.05291005	Shanghai–Suzhou	28.48540991
Hangzhou–Shaoxing	32.93650794	Nanjing–Suzhou	26.86428902
Shanghai–Hangzhou	30.82010582	Wuxi–Nanjing	25.24316813
Shanghai–Jiaxing	28.96825397	Wuxi–Shanghai	25.24316813
Shanghai–Ningbo	28.30687831	Shanghai–Changzhou	23.31326231
Hangzhou–Ningbo	26.32275132	Nanjing–Changzhou	22.85008492
Hangzhou–Jiaxing	23.41269841	Wuxi–Suzhou	22.38690752
Shanghai–Nantong	20.8994709	Suzhou–Changzhou	21.61494519
Hangzhou–Suzhou	20.8994709	Shanghai–Hangzhou	21.22896403

（4）Difference 2: Rail passenger transport has shorter average travel times

Rail passenger trips take noticeably less time than road passenger trips. Figure 6-1 shows that

HSR trips within 90 minutes account for 60% of the total, while coach trips account for less than 20%. Figure 6-2 shows that HSR trips in the YRD most commonly take 30–60 minutes, whereas coach trip times are concentrated in the 90–180 minute range, clearly longer than HSR trips.

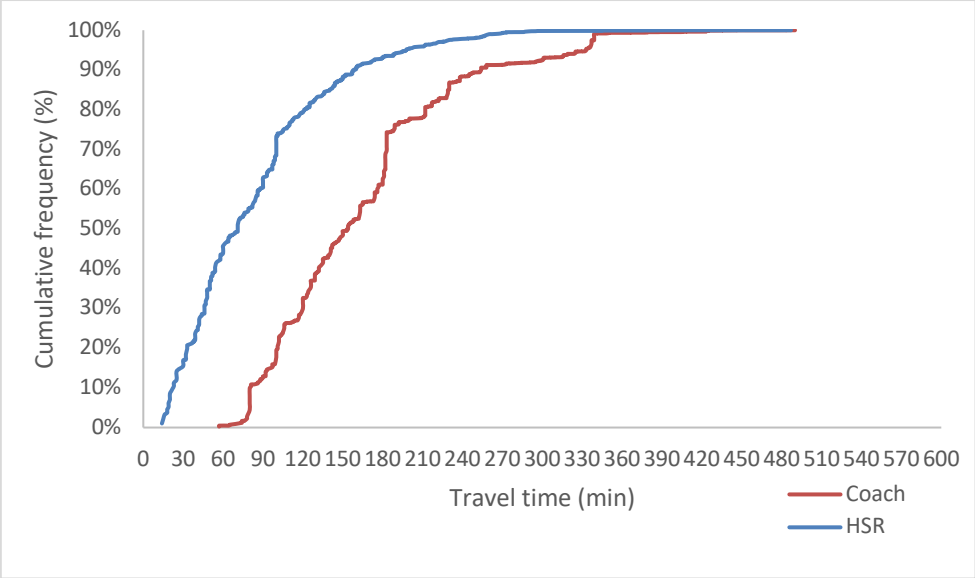


Figure 6-1 Cumulative distribution of travel time for coaches and HSR

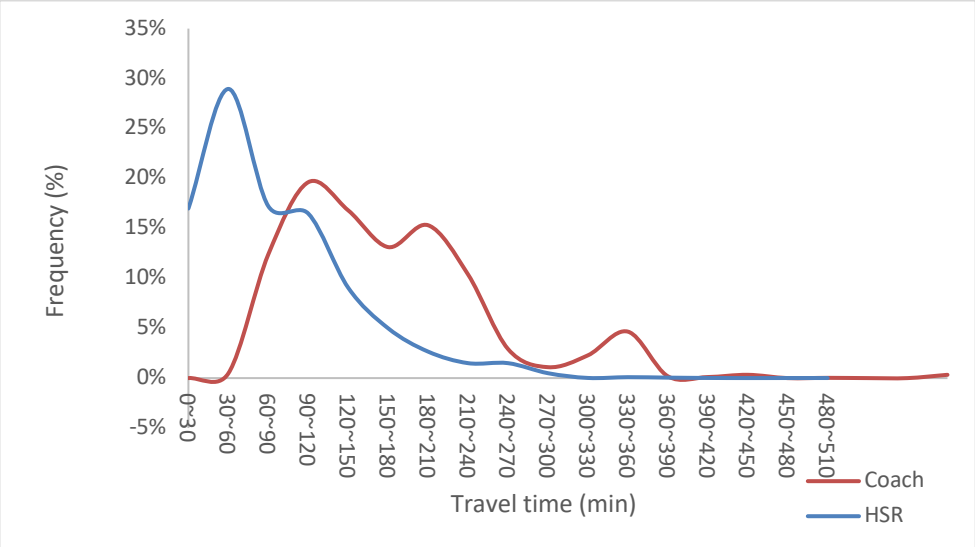


Figure 6-2 Distribution of travel time for coaches and HSR

(5) Overlapping network analysis

We analyzed the overlapping part of the road and HSR passenger networks (i.e. city pairs that have both an HSR station and a coach station and are connected by both inter-city coaches and HSR), obtaining 26 cities and 289 city pairs with 6,129 coach services and 10,661 train services, which account for 81% of all coach services and 82% of all HSR services. This overlap shows that passengers in many cities have multiple options between road and rail, creating intense competition between coach and HSR services.

3.2 Overview of the Questionnaire Data

3.2.1 Questionnaire Overview

A total of 438 questionnaires were collected. Women responded slightly more than men, but the overall ratio was close to 1:1. Respondents were mainly aged 18–25, accounting for 80% of all responses. The age structure matches the occupational composition: most respondents were full-time students (over 70%), while all other occupations were few in number.

For the question "Would you choose a long-distance coach for inter-city travel in the YRD?", about half of respondents chose "It depends", slightly more than those who chose "Definitely not". The former went on to rate various service indicators of coaches, while the latter were asked to select the mode they usually use instead of coaches; those who chose "HSR, bullet trains or intercity rail (hereafter HSR)" then rated HSR on the same items. The results show that most of those who chose "Definitely not" usually take HSR, i.e. HSR is the main substitute for long-distance coaches.

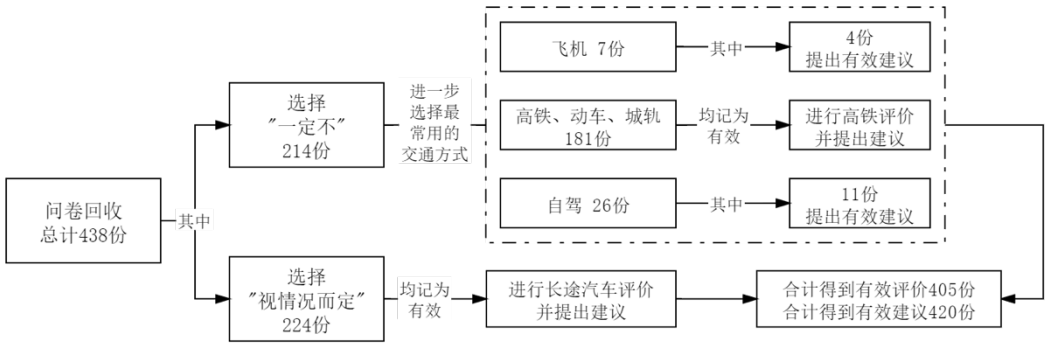
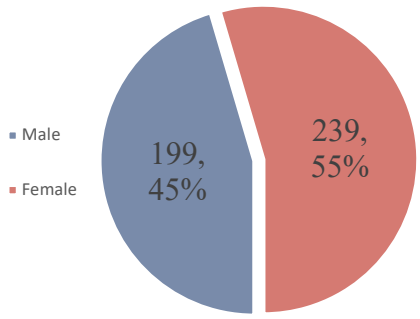


Figure 7 Questionnaire overview



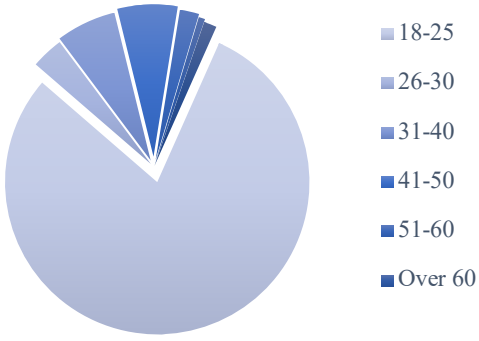
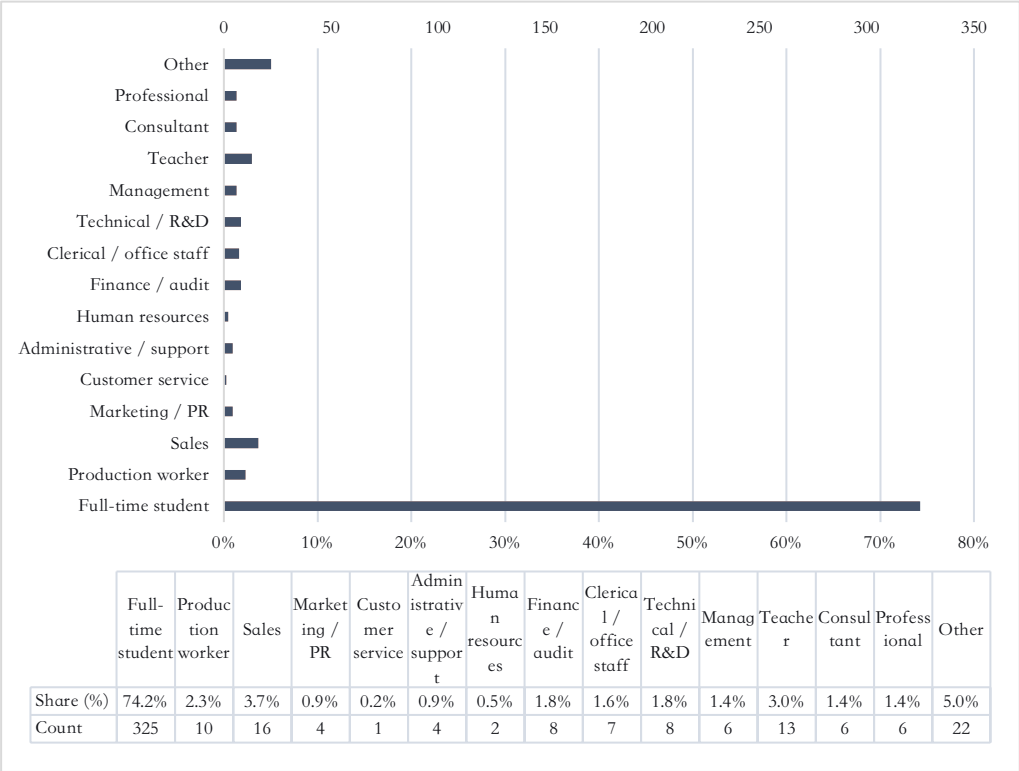


Figure 7-1 Gender of respondents Figure 7-2 Age of respondents



	Full-time student	Production worker	Sales	Marketing / PR	Customer service	Administrative / support	Human resources	Finance / audit	Clerical / office staff	Technical / R&D	Management	Teacher	Consultant	Professional	Other
Share (%)	74.2%	2.3%	3.7%	0.9%	0.2%	0.9%	0.5%	1.8%	1.6%	1.8%	1.4%	3.0%	1.4%	1.4%	5.0%
Count	325	10	16	4	1	4	2	8	7	8	6	13	6	6	22

Figure 7-3 Occupations of respondents

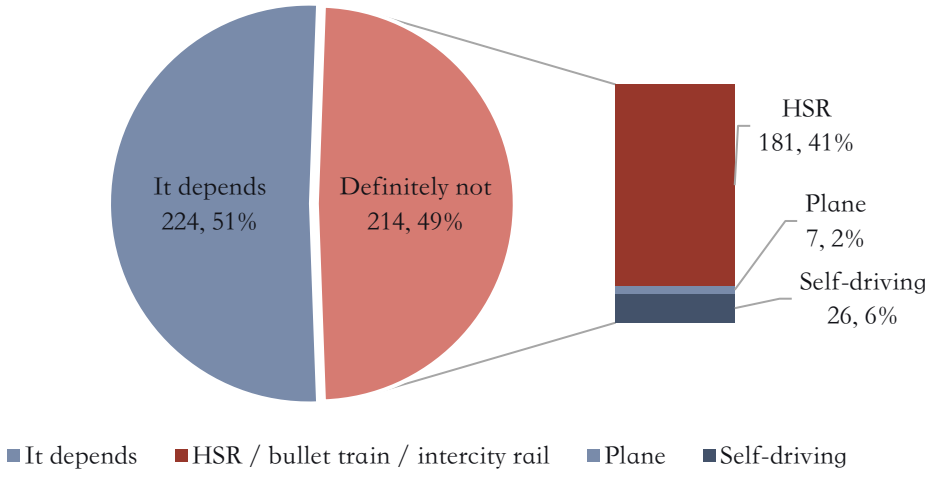


Figure 7-4 Whether respondents would choose a long-distance coach for travel

In answers to the final open-ended question, "What suggestions do you have for long-distance passenger transport in the YRD?", the public's suggestions focused on comfort and travel experience. Besides routine issues such as cabin hygiene and service quality, many passengers hoped coaches could offer special carriages, such as parent-child or women-only sections; some asked for Wi-Fi and rest facilities on board. In addition, although very few, some passengers noted the question of the future of long-distance coaches, calling for coaches to focus mainly on situations that rail cannot cover, such as cross-district trips within a city and dedicated tourist-area buses; others mentioned that the difficulties facing domestic coaches might be solved by learning from Japan's long-distance bus operating model.

3.2.2 Preliminary Analysis

Based on HSR centrality and road passenger centrality, the 27 cities were divided into four categories, and 420 valid questionnaires were selected for analysis. A centrality greater than 1 indicates that the city holds a dominant position in the YRD transport network, while a value below 1 indicates that it is marginalized.

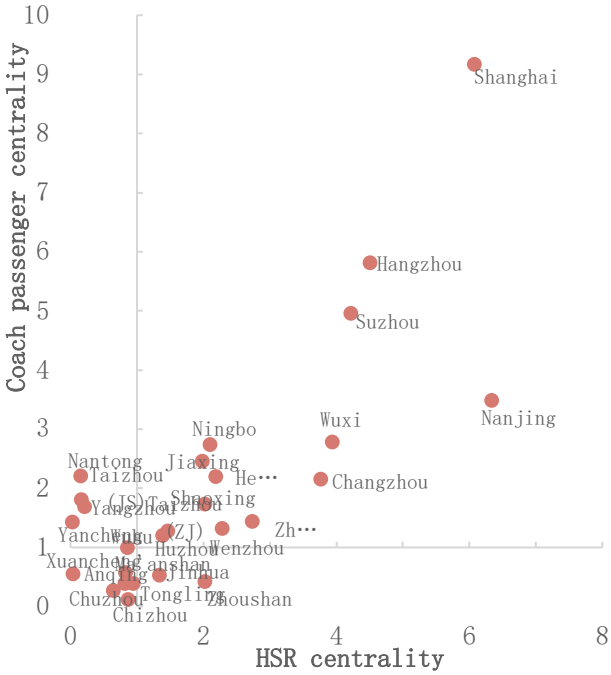


Figure 9 Criteria for the four-quadrant city classification

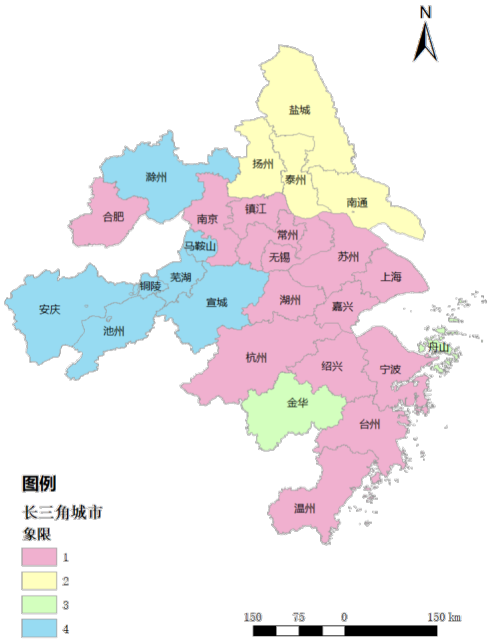


Figure 10 Results of the four-quadrant city classification

Table 4 Evaluations by city quadrant

City	Rated long-distance coaches			Rated HSR		
	Count	% of row	% of total	Count	% of row	% of total
Quadrant I cities	157	49.37%	70.09%	161	50.63%	88.95%
Quadrant II cities	41	18.30%	18.30%	7	14.58%	3.87%
Quadrant III cities	23	69.70%	10.27%	10	30.30%	5.52%
Quadrant IV cities	3	50.00%	1.34%	3	50.00%	1.66%
Total	224	55.31%	100.00%	181	44.69%	100.00%

Comparing travel-mode choices across quadrants: because HSR development is weak in Quadrant II and III cities, most respondents from these cities, especially Quadrant II cities where coach services are relatively well developed, chose "It depends" and then rated the service quality of coaches. Respondents from Quadrant I cities more often rated HSR, because although both their HSR and coach services are well developed, HSR clearly holds a higher centrality in the overall network. Quadrant IV cities are not discussed here because of their very small sample size.

3.3 Factors Influencing Passengers' Choice between Long-Distance Coaches and HSR

3.3.1 Service Quality Scores of Long-Distance Coaches and HSR

Structural equation modeling was carried out in SmartPLS. The relationships among the latent variables are shown in Figure 11, where the number on each arrow indicates the direct effect of one variable on the variable the arrow points to; the larger the number, the stronger the effect. The model yields latent variable scores representing the service quality of coaches and HSR, shown in Table 5; the reliability test results of the model are shown in Table 6.

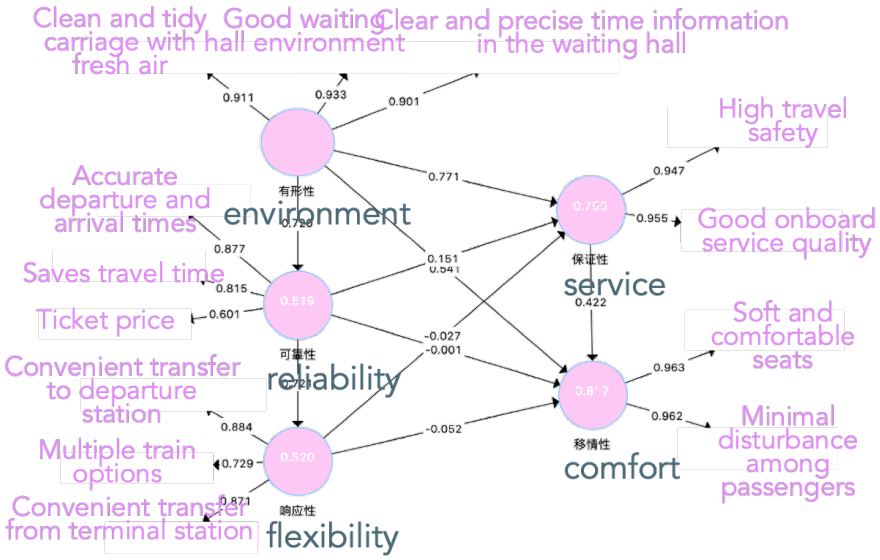


Figure 11 Overall path analysis of service quality

The path coefficients of the measurement indicators on the latent variables show that punctuality of departure and arrival has the greatest effect on reliability, while ticket price has the weakest; the waiting hall environment has the greatest effect on tangibles; transfer conditions from the origin to the departure station have the greatest effect on responsiveness; travel safety and onboard service quality have similar effects on assurance; and seat comfort and disturbance between passengers have similar effects on empathy. These results indicate that punctuality and the convenience of connections are key factors affecting transport service quality; in terms of travel experience, passengers also have high expectations for a comfortable environment and considerate service, while price plays a relatively weak role.

Table 5 Latent variable scores by dimension

Dimension	Long-distance coach	HSR
Tangibles	2.869	3.949
Reliability	3.144	4.038
Responsiveness	3.571	4.075
Assurance	2.829	3.863
Empathy	2.759	3.719

The average latent variable scores show that HSR scores higher than coaches on every service

quality dimension and is relatively balanced across them. Coaches perform particularly well on responsiveness but have major shortcomings on the other dimensions. For both modes, the highest and lowest scores occur on responsiveness and empathy, indicating that both can provide passengers with fast and effective service, but also that the personalization and thoughtfulness of passenger transport services in China need improvement. In addition, reliability ranks second among HSR's indicators and is very close to its responsiveness score, showing that HSR offers high-quality service on core travel factors such as time and price. Coaches score below 3 on tangibles and assurance, indicating considerable room for improvement in their facilities and the competence of their staff.

Table 6 Test coefficients

Latent variable	AVE	Composite Reliability	R Square	Cronbach's Alpha
Assurance	0.905	0.95	0.758	0.895
Reliability	0.598	0.813	0.519	0.664
Responsiveness	0.69	0.869	0.52	0.77
Tangibles	0.837	0.939		0.903
Empathy	0.927	0.962	0.817	0.921

Table 6 shows the calculated values of each latent variable. All five AVE values exceed 0.5, indicating that these latent factors capture a high level of distinct information from their corresponding measurement variables; Cronbach's Alpha and Composite Reliability are all greater than 0.7, indicating that the five latent variables in the model have good reliability; R Square values are within an acceptable range for all variables except responsiveness, indicating that the model fits well.

3.3.2 Analysis of Factors Influencing Passenger Choice

The questionnaire data were processed with binary logistic regression, and after several iterations the following classification table was obtained; the model's prediction accuracy is 75.6%.

Table 7 Classification table

Observed		Predicted		
		Mode choice		Percentage correct
		HSR / bullet train / intercity rail	Long-distance coach	
Mode choice	HSR / bullet train / intercity rail	132	49	72.9
	Long-distance coach	50	174	77.7
Overall percentage				75.6

Based on the variables in the final equation, with significance below 0.05 as the criterion, the factors finally identified as influencing mode choice are age group, reliability, responsiveness and assurance.

Table 8 Variables in the final equation

	B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)

								Lower	Upper
Step 4	Age group	0.408	0.124	10.910	1	0.001	1.505	1.181	1.917
	Reliability	-1.483	0.308	23.129	1	0.000	0.227	0.124	0.415
	Responsiveness	1.194	0.273	19.054	1	0.000	3.300	1.931	5.640
	Assurance	-0.877	0.212	17.078	1	0.000	0.416	0.275	0.631
	Constant	-0.607	0.314	3.743	1	0.053	0.545		

The final logistic regression equation is:

$$\text{Logit } P=-0.607+0.408X_1-1.483X_2+1.194X_3-0.877X4$$

where X_1 is age group, X_2 is reliability, X_3 is responsiveness, and X_4 is assurance.

The results show that different age groups choose travel modes differently: the older the passenger, the more likely they are to travel by coach. Responsiveness is positively correlated with choosing coaches, reflecting coaches' competitive advantage of flexibility and convenience, while reliability and assurance are negatively correlated with choosing coaches, reflecting HSR's strengths in travel quality and service attitude.

4. Conclusions and Discussion

4.1 Conclusions

The YRD coach and HSR passenger networks share two features, a distance-decay effect and a spatial agglomeration effect: both modes are concentrated on short- and medium-distance passenger transport and compete intensely in the 200–300 km range, and both are concentrated in a few major node cities and distributed along the main trunk cities. The difference is that, compared with HSR, the coach network has tighter internal connections. From the perspective of passengers' travel intentions, because rail transport in YRD cities is well developed and road passenger transport offers poor comfort and experience, nearly half of respondents are unwilling to take inter-city coaches. The punctual and efficient operation of coaches is the core competitive strength that keeps them a major force in today's passenger market, while the flexible location of coach stations and the frequent and varied schedules allow them to retain a certain market share, especially among older passengers.

4.2 Recommendations

To secure a place in the HSR era, long-distance coaches must identify their target cities and market segments. At present, the main markets for coaches are cities where road passenger infrastructure is better than or comparable to rail, such as Jinhua and Yancheng; the target group is mainly older passengers, whose service needs deserve more attention, while appropriate measures can also be taken to expand the youth market.

Specific improvement strategies can start from the following aspects:

(1) Reliability: raise coach operating speeds through technological upgrades and improve route forecasting to increase the punctuality of departures and arrivals;

(2) Assurance: strengthen driver training and qualification supervision, and cultivate a personalized and considerate service culture among crew members;

(3) Develop new products to attract young people: coach stations and coach companies can

attract young people to travel by coach by increasing promotion aimed at them and offering services they favor (e.g. combining coach travel with tourism).

4.3 Future Research

The main limitation of this study is that coach ticketing data are not yet fully available online, so the scraped data contain some errors. Obtaining actual ticketing data from transport authorities or private companies in the study area would be an effective way to further improve data accuracy; in addition, studying the coach operating models of other regions such as Europe and Japan could help identify concrete pathways for optimizing domestic coach services and resolving their current difficulties.

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